Accuracy, Calibration and Script Robustness of Decision Models on Bengali: A Zero-Shot Benchmark with Hand-Written Romanization

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Abstract

Decision models answer a typed question in one forward pass and return a probability for every option. I tested one hosted model, Jev, and eight open ones on five Bengali tasks, zero-shot, in Bengali script and in Latin letters. Jev scored highest on all five tasks among the models run on the full test sets; on Belebele it reached 0.918 (95% CI 0.900 to 0.936). The open d3-flash matched or beat it on three tasks, on a subset of about 405 items per task. Romanization hurt every model that was above chance on Bengali script, and tool-made romanization hurt more than real typing. On 877 items that I rewrote by hand in Banglish, Jev lost 11.2 points against the Bengali original, where rule-based romanization of the same text cost it 19.4. The model ranking was the same either way (Kendall $\tau = 0.89$ over 9 models), so synthetic romanization ranks models well but overstates how much they lose. Every open model was overconfident on Bengali text, and one temperature fitted on a separate development set removed most of that, but less of it on romanized text. Code, tables and the hand-written Banglish are at `https://github.com/ihpolash/system-one-testing-for-bengali`.

1 Introduction

A decision model reads a text and a typed question and returns one probability per option, without generating anything. That makes the output easy to act on: a pipeline can accept confident answers and send the rest to a larger model or a person [4, 16]. Nearly all of the evidence for this comes from English. The one study that left English shows why that matters. On Chinese tasks Laya-multilingual scored 40.2% where Jev scored 68.4% [19]. For Bengali I found only the per-language Belebele and SIB-200 numbers that ship with the Jev benchmark [4]. I found no study that asked whether a decision model's confidence still means something in Bengali.

Bengali has a second difficulty. A great deal of everyday Bengali is typed in Latin letters, with no standard spelling. Work on other Indic languages and on Bangla dialects tests romanized text produced by a transliteration tool, and finds large losses [11, 12]. A tool applies one rule per letter. People do not. I could not find a measurement of how far tool-made romanization differs from what people actually type, in terms of what a model gets right.

I made that measurement for Bengali. I rewrote 935 sentences from three datasets in Banglish by hand and scored all models on the same items in four spellings (Section 4.2). Around it sits a zero-shot benchmark of nine decision models on five tasks, with calibration, risk–coverage and option-order analyses (Section 4). Every number here is produced by the released code, and the

Study	Decision models	Bengali	Romanized	Calibration	Hand-written text
Deußer et al. [4]	Jev only	per-language file	Belebele, SIB-200	yes	no
Wang et al. [19]	trained model	no	no	yes	no
Khullar et al. [12]	no	no	tool-made	routing only	no
Jawad et al. [11]	no	dialects	tool-made	no	no
This paper	9 models	5 tasks	rule, natural-style, hand	yes	877 items

Table 1: What each closest study covers, from reading the papers. A "no" means the paper does not report it.

harness reproduces published numbers from independent groups for Jev, Laya-multilingual and laya-bangla (Section 4.5). This is not the first Bengali result for any decision model, since Jev's Belebele and SIB-200 numbers exist. Kev, d3-27B and Microsoft-Decision-1 are not covered.

2 Related work

Table 1 shows what the closest studies cover. In English, decision models have been checked for calibration [16], for annotation work with open models [9], for sensitivity to option names [18] and to option order [13], and for cost and wording robustness [17]. Other languages are thin. Chinese-Jev trains a Chinese model [19], and one paper compares Jev's cultural-value answers in English and Arabic [2]. Script robustness has been studied for generative models only [11, 12], and the Bengali half of the large romanization corpus of Gharami et al. [7] is generated by rule, not typed by people. For calibration and selective prediction I use the standard measures [6, 8]; the option-rotation test follows Zheng et al. [20]. The tasks are Belebele [3], SIB-200 [1], MASSIVE [5], SentNoB [10] and the Bengali translation of XNLI.

3 Benchmark

Models. Jev (`jev-1.13.0`) is reached through its API. The open models ran on Kaggle GPUs: Laya-multilingual, an mmBERT-based encoder [15]; laya-bangla v2, a Bangla fine-tune of it (commit `4c5eb48932`); Julia-1, with 144M parameters; and the vLLM-SR Decision 3.0 models d3-edge, d3-lite, d3-nano, d3-mini and d3-flash [14]. d3-flash has 9B parameters and did not fit comfortably, so it ran 4-bit (NF4) on a hash-ordered subset of about 405 items per task. Gemma-4-E4B-it is not a decision model; it serves as a baseline, with each option scored by the next-token probability of its label and the probabilities normalized over the options. d3-27B and Microsoft-Decision-1 (API only) were not run.

Tasks. The five tasks are in Table 2. Each item is a single-choice question with English instructions and option names; only the passage or utterance changes between conditions. I cut XNLI-bn and MASSIVE to hash-ordered subsets of 1,500 items so that any smaller run is nested in a larger one. Bengali option names changed Jev's accuracy by at most 0.8 points on SentNoB, XNLI-bn and SIB-200, so I kept English names throughout. The metric is accuracy, and macro-F1 for SentNoB.

Text conditions. *Bengali* is the dataset text. *Rule-based romanization* maps each letter to Latin letters by a fixed rule set, without diacritics. *Natural-style romanization* replaces each word with

Dataset	Task	Options	Metric	Text origin	Test items (all / used)
Belebele	reading comprehension	4 options	accuracy	human translation (FLORES-200)	900 / 900
SIB-200	topic	7 options	accuracy	human translation (FLORES-200)	204 / 204
XNLI-bn	natural language inference	3 options	accuracy	machine translation of XNLI	4,895 / 1,500
MASSIVE bn-BD	intent	59 options	accuracy	human translation	2,974 / 1,500
SentNoB	sentiment	3 options	macro-F1	native social-media comments	1,586 / 1,586

Table 2: Tasks. XNLI-bn is the machine-translated Bengali XNLI, so its text is translationese; SentNoB is the only native Bengali set. The SentNoB split is a third-party mirror whose provenance we could not verify.

its most frequent spelling in a public corpus of natural Banglish (nahidstaq/bangla-transliteration-data) and falls back to rules tuned on held-out words from that corpus. Both are synthetic. Belebele passages were not romanized.

Hand-written Banglish. I took 935 Bengali sentences from SentNoB, MASSIVE and XNLI-bn, corrected spelling errors in 81 of them, and wrote every sentence again in Banglish, the way I would type it. All of it was written by hand, one sentence at a time, from the corrected Bengali. Items are matched to the datasets by their original text. An XNLI-bn item needs both sentences, so the usable set is 877 items: 494 from MASSIVE, 325 from SentNoB and 58 XNLI-bn pairs. Spelling correction alone moved Jev by +0.2 points (95% CI -0.2 to +0.7) and no model by more than 0.6, so I compare the Banglish with the corrected Bengali, and I apply both romanizers to the corrected text as well.

Statistics. Intervals are 95% percentile bootstraps over premise or passage clusters (Belebele, XNLI-bn) or over items, and over items for the paired comparisons on the hand-written set. Calibration is measured with expected calibration error over 15 equal-width bins (ECE), the signed gap between confidence and accuracy, Brier score and risk–coverage curves. Temperature scaling is fitted on a development split disjoint from the test items, on Bengali text only, and applied unchanged. Option-order sensitivity uses three cyclic rotations of the options. I report no significance tests. Comparisons are point estimates with intervals, with no correction for the number of comparisons.

4 Results

4.1 Accuracy on Bengali script and romanized text

Table 3 and Figure 1 give the main results. Jev has the highest point estimate on every task among the models run on the full sets. The closest open model is d3-nano on XNLI-bn (0.746 against 0.751), with d3-mini at 0.729, and the intervals in Table 7 overlap. d3-flash, on its subset, scores 0.828 on XNLI-bn against Jev's 0.752, and trails on Belebele (Table 4). It is above Jev on three tasks, but with 204 to 405 items per task I read that as level. Gemma-4-E4B holds up best under romanization: on SIB-200 it keeps 0.730 of 0.853, where d3-mini keeps 0.549 of 0.838. laya-bangla beats its base model on SIB-200 (0.750 against 0.588) and on MASSIVE (0.575 against 0.179) and still sits below Jev, d3-mini and Gemma-4-E4B on every task. On SentNoB, Laya-multilingual (0.263) and Julia-1 (0.403) score below the majority class (0.412). Julia-1 cannot take the 59-option MASSIVE menu at all. Laya-multilingual's SentNoB score also depends on the wording of the question: macro-F1 is 0.263 with my default instruction, 0.321 with option names only, and 0.376 with Bengali wording. I report the default and flag the range.

Model	Belebele	SIB-200	XNLI-bn	MASSIVE	SentNoB
Jev	0.918	0.897 (0.725)	0.751 (0.572)	0.757 (0.481)	0.623 (0.527)
Gemma-4-E4B	0.773	0.853 (0.730)	0.631 (0.519)	0.689 (0.429)	0.590 (0.533)
d3-mini	0.727	0.838 (0.549)	0.729 (0.562)	0.749 (0.383)	0.579 (0.486)
d3-nano	0.702	0.848 (0.377)	0.746 (0.497)	0.705 (0.221)	0.592 (0.408)
d3-lite	0.577	0.760 (0.265)	0.704 (0.463)	0.624 (0.121)	0.535 (0.343)
d3-edge	0.409	0.475 (0.235)	0.523 (0.445)	0.243 (0.044)	0.472 (0.366)
laya-bangla	0.313	0.750 (0.250)	0.625 (0.435)	0.575 (0.077)	0.464 (0.216)
Laya-multilingual	0.287	0.588 (0.275)	0.635 (0.383)	0.179 (0.045)	0.263 (0.141)
Julia-1	0.287	0.520 (0.284)	0.382 (0.316)	–	0.403 (0.299)
Majority class / chance	0.25	0.250	0.342	0.071	0.412

Table 3: Accuracy on Bengali script, and in brackets on rule-based romanization, rotation 0 (SentNoB: macro-F1). Belebele passages were not romanized. Julia-1 cannot take the 59 options of MASSIVE. Intervals are in Table 7.

On the d3-flash subset	Belebele	SIB-200	XNLI-bn	MASSIVE	SentNoB
Jev	0.840	0.897 (0.725)	0.752 (0.557)	0.799 (0.519)	0.637 (0.577)
d3-flash (4-bit)	0.730	0.892 (0.627)	0.828 (0.598)	0.819 (0.412)	0.653 (0.519)

Table 4: d3-flash (4-bit) and Jev on the same hash-ordered subset (204 to 405 items per task); romanized in brackets. †4-bit quantization of the 9B checkpoint.

4.2 Hand-written Banglish

Figure 2 and Table 5 put four spellings of the same 877 items side by side. Every model above chance does better on my Banglish than on rule-based romanization, by 8.2 to 17.1 points. For Jev the gain is +8.2 points (95% CI +5.1 to +11.1). Natural-style romanization lands between the two, so tuning a romanizer on real typing narrows the gap but does not close it. The gain varies by task (Figure 3). For Jev it is +13.6 points on MASSIVE (+9.7 to +17.4), +8.6 on XNLI-bn (+0.0 to +17.2, from only 58 pairs), and +0.0 on SentNoB (-4.9 to +4.6). My guess is that MASSIVE's short commands, full of English loan words that people spell the same way every time, are what a tool gets wrong. I did not test that. Julia-1, near chance on these items, shows no gain.

The order of models barely moves. Kendall τ between the hand-written ordering and the rule-based one is 0.89, and 0.94 for natural-style, over 9 models scored on the same items. That is one comparison, not eight independent ones.

4.3 Calibration and selective prediction

Overconfidence differs a lot between models. Raw ECE on Bengali script, averaged over XNLI-bn, MASSIVE and SentNoB, runs from 0.074 for d3-mini to 0.432 for Julia-1 (Table 6). One temperature, fitted on a separate Bengali development split, lowers ECE for every model on Bengali text; for Julia-1 on XNLI-bn it takes ECE from 0.420 to 0.098. On romanized text the same temperature leaves a higher ECE for eight of nine models. d3-mini is the exception. Jev's scaled ECE is 0.095 on romanized text against 0.047 on Bengali, and laya-bangla's is 0.172 against 0.034. Figure 4 shows the reliability curves and Figure 5 the risk–coverage curves. Confidence still ranks answers within each condition, but romanization lowers accuracy at every coverage level, including among the most confident answers.

A confidence gate set on Bengali development text to reach 90% accuracy does not carry over.

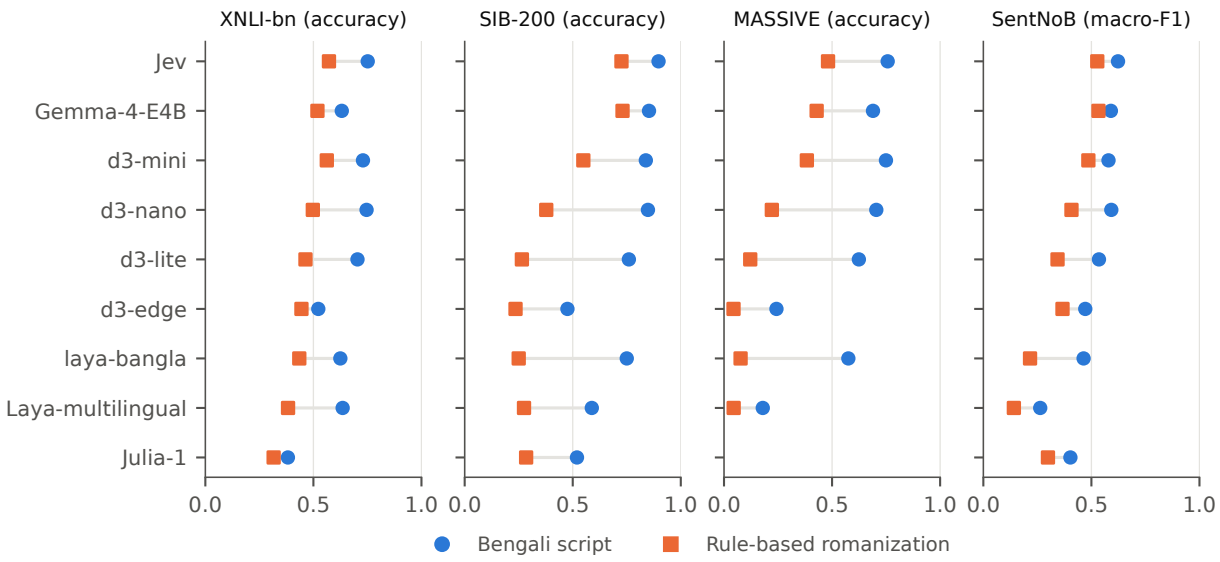


Figure 1: Bengali script against rule-based romanization for each model and task. Lines join the two conditions for the same model.

On MASSIVE, Jev accepts 69% of Bengali items at 91% accuracy, but only 23% of romanized items, at 88%. d3-mini accepts 68% at 90% and 22% at 83%. For laya-bangla the accepted romanized items are 17% accurate, at 1% coverage. So for Jev and d3-mini the gate keeps its accuracy by refusing most romanized input, and for laya-bangla it fails. This rests on one task and, for some models, on few accepted items, so I treat it as exploratory.

4.4 Option-order sensitivity

Across three rotations of the options, Jev gives the same answer 93% of the time, d3-nano 89%, d3-lite 87% and Gemma-4-E4B 81%. The encoders are far less stable: laya-bangla 67%, Laya-multilingual 53%, Julia-1 35% (Figure 6; per task in Table 8).

4.5 Checks on the harness

Before trusting any of this I tried to reproduce other people's numbers. From the Jev authors' released responses, Belebele ben_Beng accuracy is 0.9167; my own run gives 0.918. For Laya-multilingual I get MASSIVE 0.179 and SIB-200 0.588, against 0.18 and 0.57 to 0.58 reported by the laya-bangla authors. For laya-bangla I get SIB-200 0.750 against their 0.76. Resending 300 Jev requests gave the same answer 99% of the time, and a newer Laya package version changed no answer on 6,563 requests. One comparison I could not make: laya-bangla's authors report MASSIVE 0.81 with a two-step scenario-then-intent protocol, and I ask one 59-way question and get 0.575. The study used 92,618 Jev requests, about 50.9M input tokens and $2.14 at $0.042 per million, plus about 26 GPU hours on Kaggle. I measured latency under mixed conditions, so it is not comparable across models, and I do not report it.

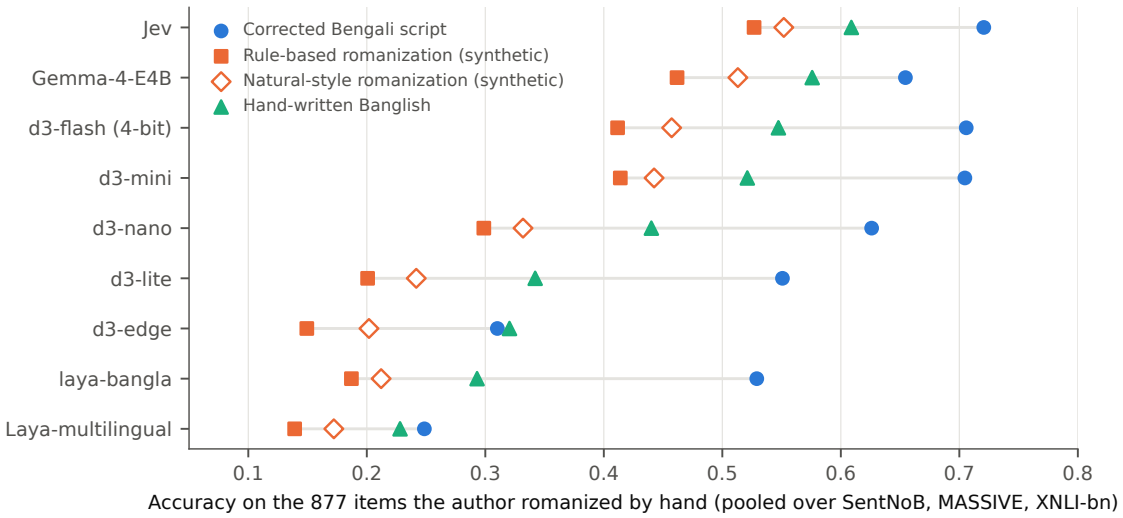


Figure 2: Accuracy of each model on the same 877 items in four spellings.

	Accuracy				Change in points (95% CI)	
Model	Bengali	Rule-based	Natural-style	Banglish	Banglish − Bengali	Banglish − rule-based
Jev	0.721	0.527	0.552	0.609	-11.2 [-14.1, -8.2]	+8.2 [+5.1, +11.1]
d3-flash (4-bit)†	0.706	0.412	0.457	0.547	-15.8 [-19.3, -12.7]	+13.6 [+10.7, +16.8]
Gemma-4-E4B	0.655	0.462	0.513	0.576	-7.9 [-10.7, -5.1]	+11.4 [+8.8, +14.3]
d3-mini	0.705	0.414	0.442	0.521	-18.4 [-22.0, -14.9]	+10.7 [+8.0, +13.6]
d3-nano	0.626	0.299	0.332	0.440	-18.6 [-22.4, -15.1]	+14.1 [+11.3, +17.4]
d3-lite	0.551	0.201	0.242	0.342	-20.9 [-24.2, -17.1]	+14.1 [+11.3, +17.2]
d3-edge	0.310	0.149	0.202	0.320	+1.0 [-2.7, +4.7]	+17.1 [+14.1, +19.7]
laya-bangla	0.529	0.187	0.212	0.293	-23.6 [-27.3, -20.0]	+10.6 [+8.0, +13.3]
Laya-multilingual	0.249	0.139	0.172	0.228	-2.1 [-5.0, +0.9]	+8.9 [+6.6, +11.2]
Julia-1‡	0.428	0.441	0.418	0.384	-4.4 [-11.0, +2.1]	-5.7 [-11.2, +0.0]

Table 5: Accuracy on the 877 hand-written items (pooled over MASSIVE, SentNoB and XNLI-bn pairs; accuracy for all three) and paired differences with 95% bootstrap intervals over items. †d3-flash is 4-bit. ‡Julia-1 answers only the 383 items it can take.

5 Discussion

What a synthetic romanization can and cannot tell you. A romanization tool ranks models correctly and overstates their loss. For Jev, planning from tool-made test data would mean expecting 19.4 points lost on Banglish input, about 8 points too many. I do not know why people's Banglish is easier. Two explanations fit: people spell common words the way models have seen them, and my corrections removed spelling errors that a tool copies through. The control rules out only the second. Correcting the original Bengali moved no model by more than 0.6 points.

Choosing a model. On Bengali script d3-mini, d3-nano and Gemma-4-E4B come within a few points of Jev on several tasks, so the hosted model's advantage here is modest. Gemma-4-E4B is the most robust open model to romanization, and also among the worst calibrated before scaling (ECE 0.303), which temperature scaling mostly removes. The encoders need care. Their answers change with option order, and Laya-multilingual's SentNoB score moved by more than ten points

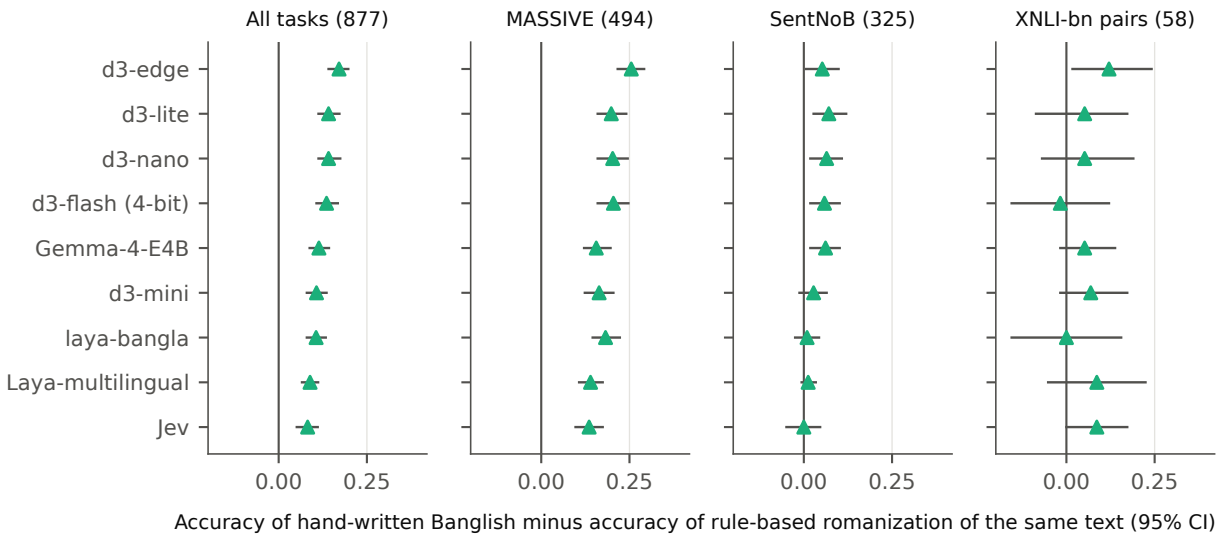


Figure 3: Accuracy of hand-written Banglish minus rule-based romanization of the same text, pooled and by task. Bars are 95% intervals.

	Bengali script		Rule-based romanization	
Model	ECE raw	ECE scaled	ECE raw	ECE scaled
Jev	0.126	0.047	0.146	0.095
Gemma-4-E4B	0.303	0.073	0.391	0.082
d3-mini	0.074	0.036	0.079	0.035
d3-nano	0.083	0.041	0.122	0.054
d3-lite	0.076	0.040	0.172	0.108
d3-edge	0.174	0.044	0.202	0.085
laya-bangla	0.204	0.034	0.331	0.172
Laya-multilingual	0.347	0.080	0.530	0.209
Julia-1	0.432	0.104	0.494	0.154

Table 6: ECE (15 bins) on Bengali script and rule-based romanization, before and after one temperature fitted on a native-text development split. Mean over XNLI-bn, MASSIVE and SentNoB.

with the wording of the question.

6 Limitations

I wrote all of the Banglish myself, so the gap is the gap for one typing style. A second typist on a subset would show how much it varies; I have not done that. MASSIVE supplies 56% of the hand-written items and XNLI-bn only 58 pairs. XNLI-bn and MASSIVE are translations, so SentNoB is the only native Bengali text, and on SentNoB my Banglish is no easier than rule-based romanization. I do not know whether that comes from social-media text or from my sample. d3-flash ran 4-bit on a subset, d3-27B was not run, and Julia-1 cannot take MASSIVE. Contamination is unknown for Jev, Gemma and the Qwen-based d3 models, and laya-bangla was trained on MASSIVE and SIB-200 training splits. The SentNoB split I used is a third-party mirror whose provenance I could not verify. I use one prompt format per model family, and Laya's wording sensitivity shows that this can move a result. The calibration and gate results are exploratory and use no multiplicity

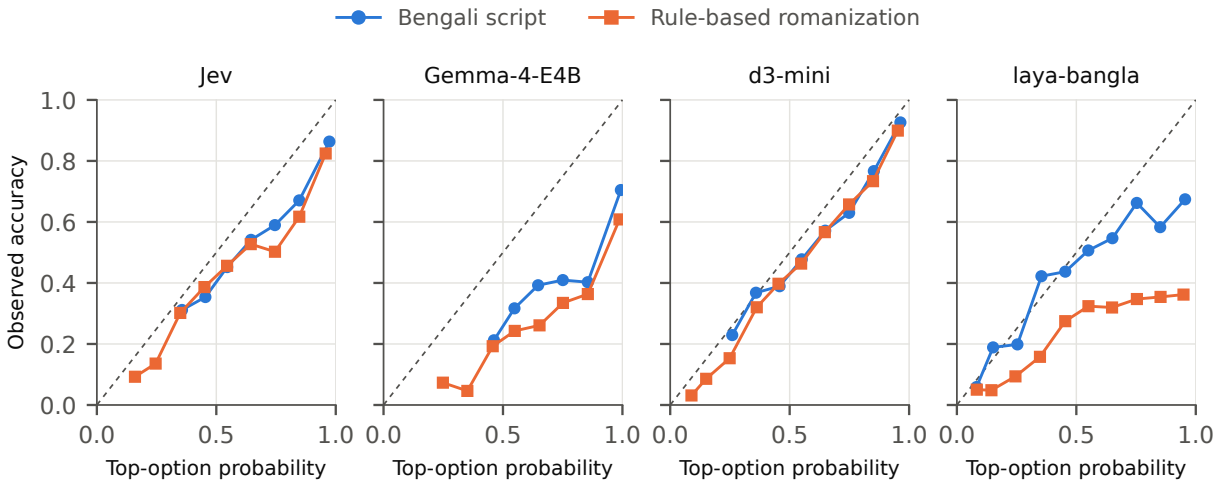


Figure 4: Reliability of four models on MASSIVE, XNLI-bn and SentNoB pooled. The dashed line is perfect calibration.

correction.

7 Conclusion

On five Bengali tasks Jev leads, several small open models come close on Bengali script, and the open models' overconfidence is largely fixed by one temperature on Bengali text but not on romanized text. Tool-made romanization is a fair way to rank models on Banglish input and a poor way to size their loss. To size it, test on text people wrote.

Availability. The repository is at `https://github.com/ihpolash/system-one-testing-for-bengali`. It holds the harness, the analysis code that generates every table and number in this paper, the summary result tables, and the hand-written Banglish for the MASSIVE and XNLI-bn sentences. SentNoB is distributed under a no-derivatives licence, so the repository contains neither its text nor the Banglish of its sentences. Item-level predictions and Jev's raw responses (provider terms) are not included.

Declaration on the use of generative AI

The author used an AI assistant (Claude, Anthropic) for implementation support, analysis and figure scripting, and for drafting and editing this manuscript. All of that work was carried out under the author's direction and review. The research questions, benchmark design, evaluation protocol and the interpretation of the results are the author's. All numerical results in the tables and text are generated by the released analysis code from the stored results, and every reference was verified by the author against its primary source. The author takes full responsibility for all content of this paper, irrespective of how it was drafted.

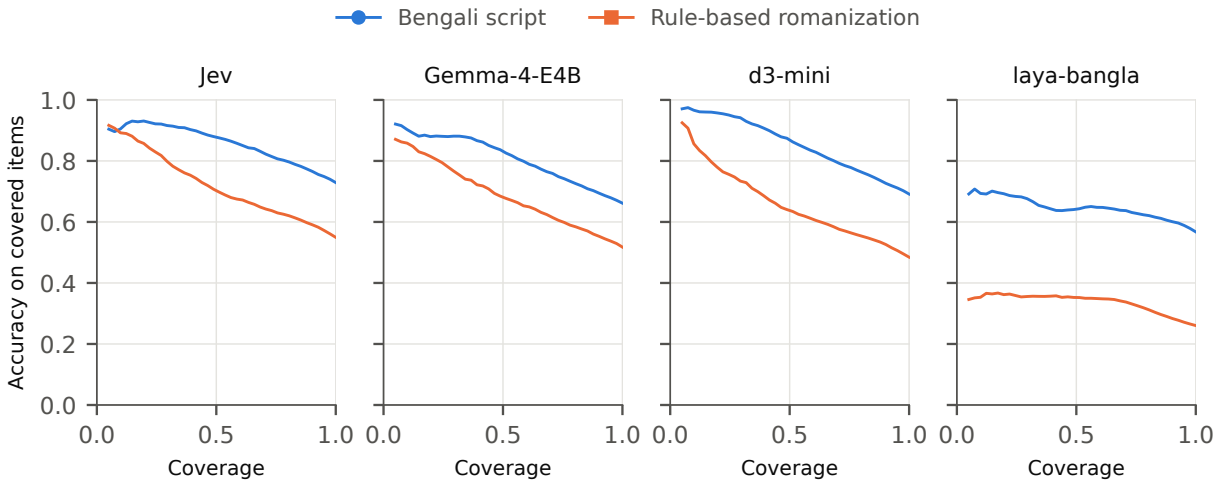


Figure 5: Accuracy on the covered items as coverage grows from the most confident answers.

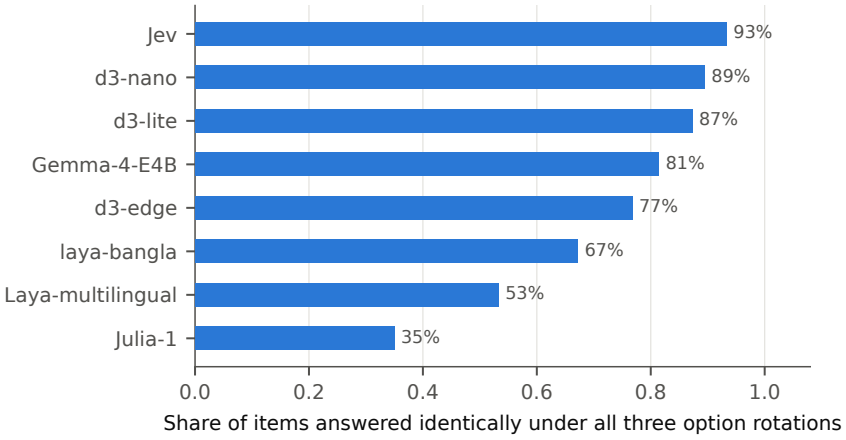


Figure 6: Share of items answered identically under all three option rotations, over all tasks.

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A Additional tables

Model	Belebele	SIB-200	XNLI-bn	MASSIVE	SentNoB
Jev	0.918 [0.900, 0.936]	0.897 [0.858, 0.936]	0.751 [0.729, 0.772]	0.757 [0.735, 0.778]	0.623 [0.603, 0.646]
Gemma-4-E4B	0.773 [0.746, 0.801]	0.853 [0.804, 0.902]	0.631 [0.608, 0.655]	0.689 [0.664, 0.713]	0.590 [0.569, 0.614]
d3-mini	0.727 [0.697, 0.757]	0.838 [0.789, 0.887]	0.729 [0.707, 0.751]	0.749 [0.727, 0.769]	0.579 [0.556, 0.604]
d3-nano	0.702 [0.672, 0.731]	0.848 [0.799, 0.892]	0.746 [0.724, 0.768]	0.705 [0.682, 0.729]	0.592 [0.566, 0.615]
d3-lite	0.577 [0.545, 0.609]	0.760 [0.701, 0.819]	0.704 [0.680, 0.727]	0.624 [0.602, 0.647]	0.535 [0.511, 0.558]
d3-edge	0.409 [0.375, 0.438]	0.475 [0.407, 0.544]	0.523 [0.499, 0.547]	0.243 [0.221, 0.264]	0.472 [0.447, 0.497]
laya-bangla	0.313 [0.283, 0.344]	0.750 [0.691, 0.809]	0.625 [0.598, 0.651]	0.575 [0.549, 0.599]	0.464 [0.440, 0.490]
Laya-multilingual	0.287 [0.259, 0.316]	0.588 [0.524, 0.652]	0.635 [0.608, 0.660]	0.179 [0.159, 0.198]	0.263 [0.245, 0.284]
Julia-1	0.287 [0.257, 0.317]	0.520 [0.456, 0.588]	0.382 [0.359, 0.406]	–	0.403 [0.378, 0.425]

Table 7: Accuracy on Bengali script with 95% cluster-bootstrap intervals (SentNoB: macro-F1).

Model	Belebele	SIB-200	XNLI-bn	MASSIVE	SentNoB
Jev	0.9 / 93%	1.0 / 97%	0.5 / 97%	1.7 / 89%	0.9 / 95%
Gemma-4-E4B	1.4 / 74%	0.5 / 93%	3.3 / 86%	4.3 / 70%	0.1 / 94%
d3-nano	1.1 / 82%	0.5 / 96%	0.7 / 96%	1.1 / 85%	0.4 / 95%
d3-lite	1.2 / 82%	1.0 / 96%	0.7 / 96%	0.5 / 80%	1.2 / 92%
d3-edge	0.3 / 75%	4.4 / 83%	1.9 / 92%	3.3 / 49%	1.4 / 90%
laya-bangla	2.1 / 28%	2.0 / 94%	1.2 / 89%	0.3 / 66%	2.8 / 88%
Laya-multilingual	1.1 / 15%	1.0 / 84%	0.1 / 91%	8.7 / 29%	4.6 / 80%
Julia-1	1.8 / 31%	2.9 / 74%	1.9 / 24%	–	10.8 / 45%

Table 8: Spread of accuracy across the three option rotations (points) / share of items answered identically under all three.